

TITLE:

Offline Facial Recognition & Liveness Detection System

Secure Workforce Authentication for Remote Location

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“Reliable Authentication Beyond Internet Boundaries”

Problem

The Problem We Identified:

In industries such as construction, highway maintenance, and field operations, workers are often located in remote areas where internet access is unstable or unavailable.

Because of this, many existing attendance systems face issues such as:

- > Dependence on internet connectivity
- > Proxy or fake attendance
- > Delayed synchronization
- > Poor performance on low-end devices
- > Privacy concerns due to cloud-based storage

These challenges reduce reliability and create operational difficulties for organizations managing large field teams.

SOLUTION

To address these challenges, we designed a mobile-based offline facial recognition and liveness detection system that can securely authenticate workers without requiring an active internet connection.

The system is capable of:

- > > Detecting and recognizing faces in real time
- > > Performing liveness verification to prevent spoofing
- > > Storing attendance records locally in encrypted format
- > > Synchronizing data to the cloud whenever connectivity becomes available

The application is lightweight, fast, and optimized for mid-range mobile devices.

INNOVATION

GPS-Based Location Verification

- Confirms attendance only at the assigned worksite location.
- Prevents fake or proxy attendance.

Auto Brightness Adjustment

- Adjusts camera brightness based on lighting conditions.
- Improves face detection accuracy outdoors.

Auto Deletion of Attendance Images

- Deletes attendance images after verification.
- Protects privacy and saves storage space.

GITHUB REPOSITORY LINK

<https://github.com/sanjaygp-22/NHAI-offline-facial-recognition.git>

SYSTEM ARCHITECTURE

The system is divided into four main layers:

Mobile Application Layer

- React Native
- TypeScript

AI Processing Layer

- Face Detection
- Face Recognition
- Liveness Verification

Technologies:

- > BlazeFace
- > MobileFaceNet
- > TensorFlow Lite
- > OpenCV

Local Storage Layer

- Offline Attendance
- Secure Local Data

Technologies:

- > SQLite
- > MMKV Storage

Cloud Synchronization Layer

- Sync After Internet Return

> Services:

- > AWS Lambda
- > AWS API Gateway
- > AWS S3

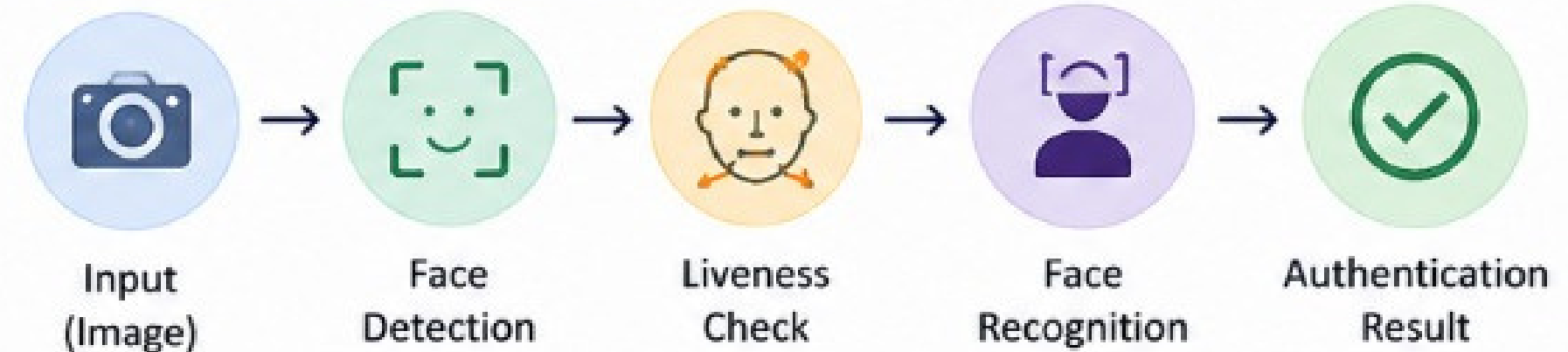
AI MODEL DESIGN

Lightweight AI Architecture

One of the main challenges in this project was building an AI system that is both accurate and lightweight enough to run completely offline on mobile devices.

To achieve this, we selected optimized models specifically designed for edge devices and real-time processing.

AI PIPELINE OVERVIEW



Entire pipeline runs offline on-device in less than 1 second.

PERFORMANCE AND EFFICIENCE

Expected Performance:

The system is designed to deliver fast and reliable authentication in real-world conditions.

Device Compatibility:

The application is optimized for:

- > > Mid-range Android devices
- > > Devices with 3GB RAM
- > > Environments without GPU support

The lightweight design ensures stable performance even in outdoor working conditions.

Performance Targets:

Metric	= Expected Result
Authentication Time	= Less than 1 second
Recognition Accuracy	= Above 95%
RAM Usage	= Below 500 MB
Total Model Size	= Less than 20 MB

SECURITY FEATURES

To ensure data privacy and protection, the system includes:

- > > AES-256 encrypted facial embeddings
- > > Secure local database storage
- > > Device-level data isolation
- > > Tamper-resistant logging
- > > Secure offline authentication handling

All sensitive data remains protected even when the device is offline

Offline Synchronization

Offline-to-Online Synchronization:

The system is designed to work continuously even when internet connectivity is unavailable.

During Offline Mode:

- > Attendance data is stored locally
- > Encrypted records are maintained securely on the device
- > When Internet is Restored
- > Data is automatically synchronized to AWS services
- > Successfully synced local records are cleared safely

Advantages:

- > Continuous operation in remote areas
- > Reliable attendance tracking
- > Reduced dependency on network availability
- > Better operational flexibility

Innovation & Real-World Impact

What makes this solution different is its focus on real-world deployment in remote environments.

Key Innovations:

- > Fully offline facial authentication
- > Lightweight AI models optimized for mobile devices
- > Real-time liveness detection
- > Secure local-first architecture
- > Offline-to-online synchronization support
- > Potential Use Cases

This solution can be implemented in:

- > Construction sites
- > Highway maintenance projects
- > Mining industries
- > Warehouses
- > Field service operations
- > Real-World Benefits
- > Reduces attendance fraud
- > Improves workforce transparency
- > Enables secure authentication in low-network regions

Conclusion

This project demonstrates how lightweight AI and edge computing can be used to build a secure and reliable offline authentication system for remote workforce management.

The proposed solution successfully combines:

- > Facial recognition
- > Liveness detection
- > Offline functionality
- > Cloud synchronization
- > Secure data storage

into a single mobile platform optimized for practical field deployment.

Future Enhancements:

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In the future, the system can be extended with:

- > GPS-based attendance verification
- > Multi-face attendance support
- > Voice authentication
- > Advanced anti-spoof AI models
- > Edge TPU acceleration

Future Scope

Thank You